What Do Document Forgery Localizers Learn from Box Masks? Ground-Truth Geometry, a Hidden Marker Shortcut, and Box Detection on AI-Inpainted Financial Documents

Harun Benli
AIC515 Image Processing
Topkapı University
İstanbul, Türkiye

***Abstract*—Benchmarks for AI-inpainted document forgery, such as AIForge-Doc, annotate each edit with the bounding box of the rewritten field. We ask what such labels teach a localizer. Using the authentic source images released with the training forgeries, we derive pixel-change and glyph masks and train the same network on box, pixel-change and glyph masks, changing nothing else. Under its own box metric the box-trained U-Net scores 0.92 pixel AUPRC; against the pixels that actually changed the same predictions score 0.37, and each ground truth ranks its own model first. A second decoder (DeepLabV3+) reproduces the effect. We find that a 1–3 px residue of the green edit marker drawn by the generation pipeline survives in over 80 % of Gemini forgeries: removing it lowers the box model's per-image AP from 0.99 to 0.77, and pasting it alone onto an authentic image is enough to light up the field. Retraining the box model on marker-repaired images (three seeds) removes this dependence while keeping most of the box score (0.91 validation AUPRC). An image-level ResNet-18 patch classifier (three seeds; 0.98 test AUROC) depends on the residue even more: repairing it lowers the detection rate from 0.995 to 0.47, and SHAP attributions concentrate on the residue pixels. A tiled YOLO11 detector trained on the same boxes is the best model under the box metric (0.97 test AUPRC), depends less on the marker and transfers better to Ideogram edits (0.87 vs. 0.63 AP), yet reaches only 0.37 validation AUPRC under the pixel-change truth. On the marker-free AIForge-Doc v2 (GPT-Image-2), without retraining, the box U-Net falls from 0.95 to 0.52 AUPRC while the detector keeps 0.89 and the marker-repaired U-Net 0.71; most of the U-Net's loss occurs on specifications whose v1 image carried the marker. Box scores on this benchmark therefore measure field detection, not pixel-level tamper localization; we recommend reporting both and auditing generation residue, and we release the code that derives our masks, controls and probes.**

***Index Terms*—document forgery, image forensics, tamper localization, ground truth, shortcut learning, AI inpainting, object detection**

I. Introduction

Image-editing services built on multimodal generators can now rewrite a single amount on a receipt or a field on a form without a visible seam. Detectors trained on classical splicing or on synthetic document tampering transfer poorly to this setting: on AIForge-Doc v1 [1], the generic localizer TruFor [2] reaches a pixel IoU of 0.36 zero-shot and document-specific models fall further, and on its successor humans perform at chance [3]. The obvious response is to train localizers on these benchmarks. We ask a prior question: *what does the ground truth of such a benchmark actually teach a localizer, and what do its scores measure?*

AIForge-Doc annotates every forgery with the bounding box of the edited field. Our audit of the released files shows that every mask is a rectangle, that only a median 49 % of the box pixels differ from the paired authentic image, and that one third of the forgeries also change pixels outside the box. A segmentation network trained on such masks is rewarded for a rectangle around any detectable cue, which is the setting in which shortcut learning thrives [4]. Annotation geometry has been questioned for natural-image edits [5], but not for AI-inpainted documents and not with controlled training. AIForge-Doc releases the authentic source of every training forgery, which lets us derive a pixel-change ground truth and change *only* the training mask.

Our contributions are:

1) **A dual-ground-truth protocol.** The same localizer trained on box, pixel-change and glyph masks (three seeds each) is scored under all three; the box-trained model's 0.92 validation AUPRC drops to 0.37 under the pixel-change truth and each ground truth ranks its own model first (Sec. V-A). The effect replicates with a second decoder (DeepLabV3+).
2) **A hidden marker shortcut, removed and re-measured.** The generation pipeline leaves a 1–3 px green line of the drawn edit marker in over 80 % of Gemini forgeries. Pasting only this line onto an authentic image makes the box model "find" the field; we retrain every condition on marker-repaired images (three seeds for the box model, one for the others) and report how much of the benchmark score survives (Sec. V-C).
3) **A region-appearance decomposition** that separates the content change from the generator's re-rendering of the

field (Sec. V-D).

4) **Box detection versus segmentation.** A tiled YOLO11 detector trained on the same boxes, scored with the same code under both ground truths and with box AP, beats every segmenter under the box metric (test AUPRC 0.969), relies less on the marker and transfers better across generators, yet scores like the box U-Net under the pixel-change truth (Sec. V-G).
5) **The shortcut in a third model family and in attributions.** An image-level ResNet-18 patch classifier (three seeds) reaches 0.978 test AUROC once its decision threshold is calibrated, yet misses half of its detections when the residue is repaired and flags 96 % of authentic images that carry only the residue; SHAP attributions concentrate on residue pixels (Secs. V-I and V-J).
6) **Negative results** on the paste-back seam, a global re-encoding trace, held-out-generator transfer with pixel supervision (three seeds), longer training and test-time enhancement.

We release the audit and the code that derives the masks, controls and probes and evaluates the models[1]; the images and the masks derived from them are not redistributed and are regenerated from the gated dataset by this code.

II. Related Work

A. Document and text tampering localization

Most document forensics work treats localization as dense prediction. DocTamper [6] paired a large synthetic corpus with a network that reads JPEG block-DCT statistics, and later localizers refine how frequency and RGB cues are fused or made robust to recompression [7]–[14]. A parallel line, originating in scene text, scores whole text instances instead of pixels: Tampered-IC13 adapts scene-text detectors so that each detected instance is also judged genuine or tampered [15], [16]. Text-centric benchmarks for receipts and documents, including T-SROIE, the ICDAR 2023 competition, RTM and a recent human-crafted real-world set, broaden the data, with labels that range from instance boxes to pixel masks [17]–[20]. Newer systems couple localization with explanation through multimodal language models or generative restoration [21]–[25]. For API-inpainted financial documents, AIForge-Doc v1 and v2 [1], [3] and related studies [26]–[30] report detectors in a zero-shot setting only. Two non-archival projects train on these data: one reports a process-twin control on v2 [31] and one a tile-level page classifier [32]; neither studies pixel-level ground truth.

B. Ground-truth semantics and dataset shortcuts in forensics

A network optimizes whatever signal its labels reward, which need not be the intended evidence [4]. In forensics this has been shown for JPEG grid alignment [8], [10], for trivial artefacts of benchmark construction [33], [34], and for frequency bias [35]. Inpainting studies report that whole-image re-decoding defeats splicing localizers while handing image-level detectors a global cue unrelated to the edit [36], [37], motivating generator-fingerprint localizers [38]; they also find that fine-tuned localizers acquire an object prior once masks are semantic [39], and that small edits remain the hardest case [40], [41]. Closest to our question, Shang et al. [5] argue that object-shaped masks label many unchanged pixels and miss real edits, so mask-only scores can rank detectors wrongly; on natural images, audits likewise question what pixel agreement measures [42] and separate the shifts behind poor generalization [43]. In segmentation, box labels are a well-known form of weak supervision that must be refined before they approximate object extent [44], [45], and metric choice should follow the structure of the target [46]. We examine the same issue for box-annotated document forgeries, where the box is also the paste-back region.

C. Detection versus segmentation for small regions

Manipulation has been localized with region-proposal detectors since RGB-N [47], and single-stage detectors [48], now available in maintained form as YOLO11 [49], predict boxes directly and are scored with COCO-style average precision [50]. When the forensic label is a field box, a detector matches the annotation geometry by construction, whereas encoder–decoder segmenters such as U-Net and DeepLabV3+ [51], [52] must spread a box label over pixels that may be unchanged. Edited fields in receipts are also small relative to the page; small objects are a recognized weakness of detectors, especially after downsampling [53], and slicing the image into overlapping tiles at native resolution is a standard remedy [54]. Multi-task models that predict boxes and masks jointly [55], and recent forgery models that couple image-level detection with pixel localization [56], suggest that box and pixel supervision can be combined. To our knowledge, no prior study compares a box detector and a segmenter on the same AI-inpainted document benchmark while varying the semantics of the ground truth used for training and scoring.

III. Dataset and Audit

AIForge-Doc v1 [1] contains 4,061 forged PNG images (3,249 train / 812 test) derived from CORD, WildReceipt, SROIE and XFUND; receipts form 89.7 % and forms 10.3 %. Each image has one edited field, produced by Gemini 2.5 Flash Image (89.6 %) or Ideogram v2 Edit (10.4 %), and a binary mask. We recomputed all statistics from the files.

Mask semantics. All masks are single axis-aligned rectangles. With $D = \max_c |I^f_c - I^a_c|$ the per-pixel difference between a forgery I^f and its authentic source I^a (3,247 of 3,249 training pairs verified by perceptual hash and pixel difference), the median fraction of box pixels with $D > 8$ is 0.49 (IQR 0.37–0.62), and 33.5 % of pairs also change pixels outside the box (Gemini 35.9 %, Ideogram 11.9 %), typically by re-rendering the whole text line. Half of the Gemini outputs carry a global $\pm$1–3 re-encoding difference that disappears at $\varepsilon \geq 4$.

[1] https://github.com/alfapars-exe/AIC515

Scale. The median edited area is 0.98 % of the image (50.7 % of images below 1 %); the median box height is 33 px while images span 0.03–35 MP, so all models run at native resolution with tiling.

The marker residue. For Gemini, the dataset paper draws a 3-px bright-green rectangle on the context crop to mark the edit region [1] (Sec. 3.4 there). A saturated-green rule ($G - \max(R, B) > 50$, $G > 110$) restricted to a 6-px ring around the box and absent from the authentic image finds residue of this line ($\geq$20 px) in 83.5 % of the 2,466 Gemini and 1.4 % of the 294 Ideogram forgeries of the training split (no false alarm on 600 authentic images). Test forgeries mostly lack a paired authentic image, so for subset analyses only we label a test image as *with residue* if any saturated-green pixel lies in the 6-px box ring (86 % of Gemini, 2 % of Ideogram forgeries over the whole dataset); model inputs are never altered by this label. Marker presence and generator are therefore strongly confounded (Cramér's $V = 0.61$), more than generator and source ($V = 0.43$; WildReceipt has no Ideogram edit).

Splits and negatives. The official test split is untouched. Training images are split by source-document groups into 2,762 train and 487 validation images with no group overlap; tool, source, area and marker distributions do not differ across the three splits (χ^2 $p \geq 0.30$). Of the test forgeries 8.9 % share a source document with training; we report the 740 leakage-free images separately. The released test "authentic" folder is not the mirror of the test forgeries (86 matches), so image-level detection is evaluated on validation pairs only. Fourteen degenerate training pairs (forged $\approx$ authentic, or all changes outside the box) are excluded from training (2,748 images remain). The dataset is gated and licensed CC BY-NC-SA 4.0; beyond the examples shown in figures, we redistribute identifiers and per-image statistics only, not images or masks.

IV. Method

A. Derived ground truths

From every verified pair we derive, besides the official BOX mask M^b: PIX, the pixel-change mask $\{D > 8\}$ cleaned by a 3×3 opening, removal of components below 8 px and a 3×3 closing; and GLYPH, Sauvola text pixels [57] of $I^f \cup I^a$ intersected with a 7×7 dilation of $\{D > 8\}$, with polarity inverted on dark backgrounds. Median IoU with the box is 0.35 (PIX) and 0.27 (GLYPH). Because the marker line differs from the authentic image, it is part of PIX (median 10 % of its positives in affected pairs); the variants PIX_{nm} and GLYPH_{nm} remove the dilated residue pixels.

B. Controls and probes

On the validation pairs we build: *benign paste-back* (box region of the authentic image up/down-sampled 2× and pasted back, i.e. compositing without content change) and *full-image resampling* [36]. Input-level probes (Fig. 1) are: *forged-repaired* (I^f with residue pixels copied from I^a); *authentic+marker* (I^a with only the residue pixels from I^f); *render-only* (I^f with the dilated PIX region and the residue reverted to I^a, so no character changes but the generator's sub-threshold re-rendering of the field remains); *content-only* (I^a with only the dilated PIX_{nm} region taken from I^f, so characters change but the surrounding re-rendering and the marker are absent); a boundary/placebo/interior blur ablation of the box; and test-time degradations (JPEG Q75, 0.75× rescale, Gaussian σ=1, CLAHE, grayscale).

C. Models

Segmenters. A U-Net [51] with an ImageNet-initialised ResNet-18 encoder [58] (14.3 M parameters) is trained on 512×512 native-resolution crops (70 % centred on training-mask positives with jitter, 30 % random), BCE+Dice loss [59], AdamW [60] (10^{-4}, cosine), batch 16, bf16, 10 epochs, model selection by validation AUPRC under the *training* mask. Inference is full-image sliding window (512, stride 448); the 44 images whose longer side exceeds 4096 px are downscaled to 4096 for inference and the map is upsampled back. The training mask is the only factor varied (BOX, PIX, GLYPH; seeds 42–44). One-factor extensions (seed 42 unless stated): (i) DeepLabV3+ [52] with the same encoder; (ii) training on marker-repaired images with BOX (seeds 42–44), PIX_{nm}, GLYPH_{nm}; (iii) a joint model with a second output channel supervised by BOX (loss $\mathcal{L}_{\text{pix}} + 0.5\,\mathcal{L}_{\text{box}}$, evaluation on the pixel channel; seeds 42–44 for the PIX variant); (iv) 20 epochs; (v) Gemini-only training for the held-out-generator test (seeds 42–44).

Box detector. YOLO11 [49] (single class) is trained on the same BOX boxes. Because the median field is 33 px high, images are cut at native resolution into 1024×1024 tiles (stride 768; every tile holding $\geq$ 50 % of the box is labelled with the clipped box, and one other random tile per image is kept with an empty label, which can contain a sliver of the field) and inference is tiled with cross-tile NMS, following slicing-aided detection [54]. Augmentation is conservative (no mosaic or scale jitter, which would add resampling traces); AdamW 10^{-3}, batch 8, 30 epochs, patience 10, checkpoint selection by Ultralytics' default fitness (0.1 mAP@0.5 + 0.9 mAP@[.5:.95]) on validation tiles. Boxes are taken from the released metadata, which agrees with the mask extent except for 12 images (0.3 %); its end coordinates are inclusive, so the labels are one pixel short in width and height. One-factor ablations (seed 42): whole image resized to 1280; default Ultralytics augmentation; one extra empty authentic tile per image; model size n/s/m; Gemini-only training. For pixel metrics, each box paints its confidence into a map (maximum over overlapping boxes), so that detector and segmenters pass through *identical* evaluation code.

Patch classifier. As a third model family we train an ImageNet-initialised ResNet-18 with a single-logit head (11.2 M parameters) on 256×256 patches cropped to 224: per training forgery two jittered patches containing the edited field, one patch at the same location of the paired authentic image and, where the image is large enough, one edit-free patch of the forged image (10,795 patches from 2,748 images; BCE, AdamW 10^{-4}, cosine, batch 32, 12 epochs, seeds 42–44). A second set of runs cuts the same patches from marker-repaired

forgeries (2,126 images affected). The image score is the maximum patch probability over a sliding window (224, stride 112); aggregation rule and decision threshold (Youden's J) are selected on the 487 validation pairs and applied unchanged to the test set. A pilot version of the same recipe uses 400 training images and one seed.

D. Evaluation and statistics

Every model is scored against BOX, PIX and GLYPH on the 487 validation images and against BOX on the 812 test images (PIX where a test pair exists, n=89). Metrics are pooled pixel AUPRC (exact, from 256-bin histograms over all pixels; on a pilot run an estimate that subsampled negatives per crop gave 0.947 where the exact value is 0.805, so no subsampling is used), per-image AP, F1/IoU at a threshold τ^* calibrated per model and ground truth on validation, and, for all models, box-level AP@0.5 and AP@[.5:.95] (COCO-style [50]; segmenter boxes from connected components at τ^*, scored by their maximum probability). Image-level discrimination compares forged with authentic validation images using the 99th-percentile score (benign controls; the maximum box confidence for the detector) or the maximum score (input probes, Tables II and III). Uncertainty is a 1,000-sample bootstrap over source-document groups [61], computed on per-image scores averaged over seeds where a model has three; contrasts within a hypothesis family are Holm-corrected [62]. Four hypotheses and their falsification criteria were fixed before the full-data runs: H1, the training ground truth changes scores and model ranking; H2, the box model relies on the box boundary or rectangle shape; H3, pixel or glyph supervision transfers better to an unseen generator and degradations; H4, forgeries carry a global trace outside the box; the marker, render/content, YOLO and joint-loss analyses were added afterwards and are reported as exploratory. Seeds 43 and 44 of the marker-repaired box model and of the joint pixel model were run last, after six predictions with falsification thresholds had been committed to the public repository; all six held.

V. Results

A. H1: the ground truth decides the ranking

Table I scores every model under every ground truth. With the U-Net, the BOX-trained model reaches a pooled AUPRC of 0.923 under BOX but 0.374 under PIX and 0.257 under GLYPH; on the official test set the same predictions give 0.950 under BOX and 0.427 under PIX (n=89 pairs). Each ground truth ranks the model trained on it first, and the BOX ranking is the reverse of the GLYPH ranking. DeepLabV3+ reproduces the pattern with nearly identical numbers (0.939 → 0.382), so the effect is a property of the supervision, not of the U-Net decoder. Under its own metric the BOX model is the best localizer in the study; under a pixel-change metric it is worse than a model that never saw a box. Most of the 0.92 is therefore earned by labelling unchanged pixels inside the rectangle.

TABLE I
POOLED PIXEL AUPRC ON VALIDATION (487 IMAGES) FOR EACH TRAINING MASK (ROWS) SCORED AGAINST EACH GROUND TRUTH (COLUMNS); MEAN±SD OVER THREE SEEDS WHERE AVAILABLE. BOLD: BEST PER COLUMN WITHIN AN ARCHITECTURE.

Model	Train	BOX	PIX	GLYPH
U-Net (3 seeds)	BOX	**0.923±.015**	0.374±.008	0.257±.007
	PIX	0.579±.014	**0.610±.012**	0.511±.010
	GLYPH	0.351±.015	0.379±.012	**0.676±.019**
DeepLabV3+	BOX	**0.939**	0.382	0.256
	PIX	0.537	**0.583**	0.432
	GLYPH	0.345	0.401	**0.549**
YOLO11s (3 seeds)	BOX	**0.965±.003**	0.368±.015	0.221±.003
U-Net, marker-repaired	BOX	0.912±.006	0.373±.013	0.265±.001
	PIX_{nm}	0.513	0.544	0.455
	GLYPH_{nm}	0.348	0.371	0.661
U-Net + box aux.	PIX	0.668±.019	0.633±.013	0.523±.009
	GLYPH	0.370	0.391	0.636
U-Net, 20 epochs	BOX	0.926	0.373	0.259
	PIX	0.568	0.594	0.488
	GLYPH	0.398	0.390	0.645

B. Shortcut and trace controls (H2, H4)

The BOX model predicts rectangles (median rectangularity 0.87 vs. 0.32 for PIX and 0.05 for GLYPH) and is indifferent to blurring the box interior (relative AUPRC change $+1.2\,\%$) while a 5-px blur of the box rim costs $-2.0\,\%$ (placebo ring 0.0 %; three seeds): it uses the field's extent more than its content, but the boundary seam is only a small cue (H2 partly supported). A ResNet-18 trained on 224-px crops that never touch the box separates forged from paired authentic images at chance (AUROC 0.516; 0.980 for box-centred crops through the same pipeline), so no global re-encoding trace is exploitable (H4 rejected). Benign paste-back and full-image resampling, which composite or resample without changing content, reduce image-level AUROC (99th-percentile score) by 1.6–3.0 points for the three-seed U-Nets and by at most 4.5 points for the other standard segmenters, but by 3.6–6.5 points for the marker-repaired models (box model: three seeds; pixel and glyph models: one). The detector, scored by its maximum box confidence, loses at most 0.3 points (AUROC 0.995 against authentic images, 0.992 and 0.994 against the two controls; seed 42), whereas the box U-Net loses 2.2–2.3 points under the same maximum-score rule. The PIX model raises far fewer false alarms on authentic images than the BOX model (12 % vs. 26 % of images with >0.1 % positive pixels at 0.5; three seeds), and the joint model of Sec. V-F 10 %.

C. The marker shortcut, and retraining without it

Fig. 1 defines the input probes; Table II reports them on the 380 validation pairs whose forgery carries marker residue ($\geq$20 px after a one-pixel dilation of the residue mask; 368 pairs without the dilation). The standard, joint and marker-repaired box U-Net rows are three-seed means; DeepLabV3+ and the repaired pixel model use seed 42. For the standard box U-Net, copying only the residue pixels back from the

TABLE II

MARKER PROBES ON 380 VALIDATION PAIRS WITH RESIDUE (U-NET BOX, PIX, +AUX AND REPAIRED BOX ROWS: MEAN OF THREE SEEDS; OTHER ROWS: SEED 42). AP: PER-IMAGE AP UNDER BOX WITH 95 % GROUP-BOOTSTRAP CI FOR THE REPAIR CONTRAST; COVER: FRACTION OF THE BOX ABOVE τ^*; AUROC: FORGED-TYPE INPUT VS. AUTHENTIC, MAXIMUM SCORE.

	forged		repaired	Δ AP	auth.+marker	
Model	AP	Cover	AP	(repair)	Cover	AUROC
U-Net BOX	0.988	0.94	0.766	−0.222	0.80	0.903
U-Net PIX	0.747	0.54	0.500	−0.247	0.42	0.858
DeepLabV3+ BOX	0.982	0.92	0.807	−0.174	0.78	0.899
U-Net+aux PIX	0.795	0.56	0.600	−0.194	0.46	0.887
repaired BOX	0.939	0.89	0.956	+0.017	0.44	0.720
repaired PIX_{nm}	0.648	0.44	0.686	+0.038	0.11	0.592

authentic image lowers per-image AP from 0.988 to 0.766 ($\Delta = -0.222$ [-0.256, -0.190]), and pasting only the residue onto the authentic image makes the model cover 80 % of the box at τ^* (AP 0.905). DeepLabV3+ and the pixel-trained U-Net show the same dependence (Δ −0.174 and −0.247). On the official test set the box model's AP is 0.972 on the 627 images with residue and 0.690 on the 185 without (0.722 on the 102 residue-free Gemini images), so the gap is not only a generator gap.

Retraining on marker-repaired images removes the dependence without destroying the benchmark score. Over three seeds the repaired box model loses nothing when the residue is removed at test time ($\Delta = +0.017$ [0.001, 0.033]); the residue alone now covers 44 % of the box instead of 80 % (11 % instead of 42 % for the pixel model, one seed). Its validation AUPRC is 0.912 ± 0.006 (0.923 ± 0.015 before) and its test AUPRC 0.940 ± 0.003 (0.950 ± 0.011); paired per image, the retraining costs 0.04 AP on both splits (validation -0.038 [-0.054, -0.021], test -0.040 [-0.052, -0.029]). Its AP on residue images falls from 0.972 to 0.914, while on residue-free Gemini images it is similar (0.762 vs. 0.722, overlapping CIs). The marker is thus a shortcut the models take when it is offered, not a prerequisite for high box scores. The repaired pixel and glyph models use one seed.

D. Re-rendering versus content

Generators re-render the whole field, so inside the box many pixels change by less than the PIX threshold. Separating the two components on all validation pairs (Table III) shows that neither alone reproduces the response to a real forgery. For the standard box U-Net (three seeds) the mean probability inside the box falls from 0.86 (forged) to 0.24 (render-only) and 0.35 (content-only), and forged-vs-authentic AUROC from 0.971 to 0.637 and 0.733: the standard models respond most strongly to the conjunction of re-rendering, changed glyphs and marker. Removing the marker from training shifts the models toward content: content-only AUROC rises to 0.886 for the box model (three seeds; 0.733 before) and to 0.834 for the pixel model (one seed; 0.681 before), while their render-only response stays lower (0.724, 0.638). Content changes, not the generator's texture, become the main cue once the shortcut is gone.



Fig. 1. Input probes on one validation pair (dashed: official box). The forged field carries a 1-px green residue of the generation marker on its top edge; *render-only* keeps the generator's re-rendering but restores the original digits; *content-only* carries the changed digits into the authentic image without the re-rendering or the marker.

TABLE III

RE-RENDERING VS. CONTENT (ALL VALIDATION PAIRS; STANDARD U-NET AND REPAIRED BOX ROWS: MEAN OF THREE SEEDS, OTHER ROWS SEED 42): MEAN PROBABILITY INSIDE THE BOX AND IMAGE-LEVEL AUROC AGAINST AUTHENTIC IMAGES.

	mean p in box			AUROC vs. authentic		
Model	forged	render	content	forged	render	content
U-Net BOX	0.86	0.24	0.35	0.971	0.637	0.733
U-Net PIX	0.41	0.05	0.15	0.942	0.554	0.681
DeepLabV3+ BOX	0.87	0.28	0.42	0.972	0.644	0.755
repaired BOX	0.82	0.35	0.58	0.957	0.724	0.886
repaired PIX_{nm}	0.34	0.10	0.29	0.869	0.638	0.834

E. Held-out generator (H3), now with three seeds

Training on Gemini only and testing on the Ideogram edits of validation and test (n=126 under BOX, 41 pairs under PIX) confirms our earlier single-seed result with three seeds (Table IV). Every Gemini-only model collapses on Ideogram; the BOX model falls from 0.909 to 0.412 per-image AP and the PIX model from 0.567 to 0.264 (both scored under BOX). The relative drop is smaller for pixel supervision, but its absolute Ideogram score is lower, and under its own ground truth the pixel model is no better on Ideogram than the box model (0.279 vs. 0.259). Pixel-change supervision does not, by itself, buy generator transfer. Because Ideogram forgeries almost never carry the marker (green residue in the box ring of 2.1 % of Ideogram vs. 85.9 % of Gemini forgeries), part of this gap may be a marker gap, which this design cannot separate: we did not train Gemini-only models on marker-repaired images.

TABLE IV
HELD-OUT GENERATOR: PER-IMAGE AP WITH 95 % GROUP-BOOTSTRAP CI, IDEOGRAM VS. SOURCE-MATCHED GEMINI (VAL+TEST, MEAN OVER THREE SEEDS).

Training	Scored	Ideogram	Gemini (matched)
All tools, BOX	BOX	0.629 [0.569, 0.692]	0.912 [0.896, 0.928]
Gemini only, BOX	BOX	0.412 [0.341, 0.481]	0.909 [0.892, 0.927]
Gemini only, PIX	BOX	0.264 [0.213, 0.313]	0.567 [0.550, 0.583]
Gemini only, BOX	PIX	0.259 [0.168, 0.350]	0.349 [0.322, 0.377]
Gemini only, PIX	PIX	0.279 [0.182, 0.382]	0.577 [0.546, 0.608]
Gemini only, GLYPH	PIX	0.182 [0.121, 0.246]	0.388 [0.363, 0.413]

F. Supervision variants, training length and robustness

Joint supervision (pixel mask plus an auxiliary box channel, three seeds) is the only variant that improves the pixel model, and it does so mainly under the box metric: validation AUPRC under BOX rises from 0.579 ± 0.014 to 0.668 ± 0.019 and test BOX AUPRC from 0.638 ± 0.015 to 0.702 ± 0.020 (paired per-image AP $+0.051$ [0.043, 0.058] and $+0.046$ [0.040, 0.052]). Under PIX the gain is small and confined to validation: 0.633 ± 0.013 against 0.610 ± 0.012 (per-image AP $+0.011$ [0.005, 0.017]), while on the 89 test pairs it is absent (0.652 ± 0.003 vs. 0.654 ± 0.016; $+0.004$ [-0.008, 0.015]). The blur penalty shrinks from $-59\,\%$ to $-46\,\%$ (Table I; blur from the robustness runs), and image-level AUROC against authentic images is 0.905 ± 0.008 against 0.892 ± 0.006. The joint model still depends on the marker ($\Delta = -0.194$, Table II). *Twenty epochs* do not change the picture: BOX 0.926 (vs. 0.923), PIX 0.594 (vs. 0.610), GLYPH 0.645 (vs. 0.676); the concern that ten epochs were too few is not supported. *Test-time preprocessing.* Converting the input to grayscale is the most damaging operation we tested, costing 47–67 % of the AUPRC of every segmenter for which it was measured (DeepLabV3+, joint and marker-repaired models) and 59 % for the detector, so colour carries evidence beyond the green line. CLAHE changes AUPRC by at most 3 %. JPEG Q75, $0.75\times$ rescaling and $\sigma=1$ blur hurt pixel and glyph models more than box models (e.g. blur $-59\,\%$ vs. $-21\,\%$ for the U-Net), and the marker-repaired models are less robust to JPEG (-32 to $-50\,\%$) than their standard counterparts.

G. Box detection versus segmentation

A tiled YOLO11s trained on the same boxes is the strongest model in the study *under the box metric* and is no better than the box U-Net under the pixel metric (Tables I and V). Over three seeds it reaches 0.965 validation and 0.969 test AUPRC under BOX (U-Net 0.923 / 0.950), test F1@τ^* 0.946 and IoU 0.918 (U-Net 0.846 / 0.789) and box AP@0.5 0.880 (U-Net 0.873 for seeds 42/43; 0.391 for seed 44, whose calibrated $\tau^*=0.95$ fragments the predicted region; marker-repaired U-Net 0.804 ± 0.005 over three seeds), while its PIX AUPRC is 0.368 (U-Net BOX model 0.374). The detector depends far less on the marker: its test AP on the 185 residue-free images is 0.878 (U-Net 0.690) and 0.886 on residue-free Gemini images (0.722). It also transfers better across generators: trained on all tools it reaches 0.872 AP on Ideogram (U-Net 0.629; both three seeds), and trained on Gemini only it keeps 0.752 (one seed; U-Net 0.412 over three seeds; Gemini 0.938 vs. 0.909). The probes (seed 42 for the detector) suggest why: removing the residue costs the detector only 0.062 AP (U-Net 0.222), whereas the *render-only* input, which keeps the generator's re-rendering but no changed glyph, is still separated from authentic images with AUROC 0.865 (U-Net 0.637). The detector has learned what a regenerated field looks like, a cue that is shared by both generators but says nothing about which pixels changed.

TABLE V
BOX DETECTOR VS. SEGMENTERS ON THE OFFICIAL TEST SET (812 IMAGES, BOX). AUPRC: POOLED PIXEL AUPRC (BOX PAINTED WITH ITS CONFIDENCE); AP: PER-IMAGE AP ON IMAGES WITHOUT / WITH MARKER RESIDUE; AP_{50}: BOX AP@0.5 (ALL IMAGES). SEEDS: YOLO11S AND U-NET 3, OTHERS 1. MS: FULL-IMAGE INFERENCE ON AN IDLE RTX 5080.

Model	AUPRC	F1@τ^*	AP no-mk	AP mk	AP_{50}	ms
U-Net BOX (3)	0.950	0.846	0.690	0.972	0.873†	81
U-Net PIX (3)	0.638	0.616	0.503	0.639	0.107	81
DeepLabV3+ BOX	0.952	0.833	0.702	0.962	0.833	122
U-Net BOX, repaired	0.939	0.797	0.701	0.920	0.805	81
YOLO11s tiles (3)	**0.969**	**0.946**	0.878	0.954	0.880	98
YOLO11n tiles	0.963	0.926	0.812	0.946	0.871	81
YOLO11m tiles	**0.969**	**0.947**	**0.882**	0.954	0.898	76
YOLO11s full 1280	0.932	0.909	0.756	0.930	**0.960**	303

† seeds 42/43; seed 44 ($\tau^*=0.95$) 0.391. Zero-shot TruFor (our run): F1@0.5 0.403.

The ablations (seed 42) are only partly as expected. Resizing whole images to 1280 instead of tiling at native resolution costs 0.037 test AUPRC and 0.122 AP on residue-free images, but *raises* box AP@0.5 from 0.88 to 0.96 (0.93 on fields below 1 % of the image): a plausible cause is that fields cut at tile borders yield partial boxes that cross-tile NMS does not merge, although the untiled detector also emits about five boxes per image, so the mechanism is not established; tiling helps coverage but not box precision here; model size matters little (n/s/m: 0.963/0.969/0.969 test AUPRC); Ultralytics' default augmentation is as good as the conservative profile (0.971), contrary to our expectation that mosaic and scale jitter would destroy forensic traces; one extra authentic tile per image changes nothing (0.971). The detector is insensitive to blur, CLAHE and rescaling ($\leq$1.4 % AUPRC) but loses 49 % under JPEG Q75 and 59 % under grayscale. Contrary to our expectation, detection was not cheaper in our implementation: full-image inference takes 76–98 ms for the tiled n/s/m detectors, 303 ms for the 1280 detector, 81 ms for the U-Net and 122 ms for DeepLabV3+ (mean 3.1 MP per image), because per-tile pre- and post-processing dominates.

H. External test on AIForge-Doc v2 (GPT-Image-2)

AIForge-Doc v2 [3] regenerates the v1 forgery specifications with GPT-Image-2 and pastes back only the edited box, so its forgeries differ from their authentic source only inside the box and carry no marker residue: no pixel outside the box differs in any of the 3,066 pairs, and our colour rule fires on a single green-tinted receipt, inside its box. We applied every model trained on v1 without retraining or recalibration (τ^*

TABLE VI

AIFORGE-DOC V2 PRIMARY SET (567 PAIRS, SOURCE DOCUMENTS UNSEEN IN TRAINING), MODELS TRAINED ON V1 ONLY. AUPRC: POOLED PIXEL AUPRC; AUC: IMAGE AUROC (TOP-1 % SCORE, FORGED VS. PAIRED AUTHENTIC). ΔAP: PER-IMAGE AP CHANGE FROM THE V1 TEST IMAGE TO THE V2 IMAGE OF THE SAME SPECIFICATION, SPLIT BY WHETHER THE V1 IMAGE CARRIED THE MARKER RESIDUE (397 / 170 PAIRS). SEEDS: ROWS MARKED (3) AVERAGE THREE SEEDS, OTHERS ONE.

	AUPRC			ΔAP (BOX)	
Model (trained on v1)	BOX	PIX	AUC	mk	no-mk
U-Net BOX (3)	0.520	0.272	0.717	−0.49	−0.17
U-Net PIX (3)	0.227	0.246	0.693	−0.40	−0.17
DeepLabV3+ BOX	0.539	0.280	0.724	−0.38	−0.14
U-Net BOX, repaired (3)	0.711	0.391	0.840	−0.19	−0.01
U-Net PIX_{nm}, repaired	0.353	**0.430**	0.826	−0.11	−0.08
YOLO11s tiles (3)	**0.887**	0.354	0.919	−0.15	−0.03
YOLO11s full 1280	0.697	0.328	0.800	−0.20	−0.08
YOLO11s Gemini-only	0.843	0.338	0.865	−0.21	−0.06
YOLO11s +authentic tiles	0.863	0.345	**0.937**	−0.19	−0.06

and image thresholds from v1 validation). Because v2 reuses v1's source documents, we fixed before scoring a leakage-free primary set: the 567 v2 pairs whose v1 counterpart is in the v1 test split and whose source document never appears in v1 training (Table VI). Image-level AUROC uses the paired authentic source of every v2 forgery.

The box U-Net, 0.950 test AUPRC on v1, falls to 0.520 ± 0.018 (three seeds) and to 0.717 image AUROC; the tiled detector keeps 0.887 ± 0.006 and 0.919. Comparing the *same* specification on v1 and v2 separates the generator change from the loss of the marker: on specifications whose v1 image carried the residue the box U-Net loses 0.49 per-image AP, on residue-free ones 0.17, and the box U-Net retrained on marker-repaired images shows no loss on residue-free specifications (-0.008, 95 % CI $[-0.07, 0.05]$; three seeds) and reaches 0.711 ± 0.007 AUPRC overall. The detector loses only 0.03 on residue-free specifications. The marker-repaired pixel and glyph models are also the best pixel and glyph localizers on v2 (PIX 0.430 vs. 0.246; GLYPH 0.446 vs. 0.304). On a marker-free benchmark from a third generator, the box U-Net's v1 advantage over the repaired model is thus reversed, and most of what it loses is tied to the marker rather than to the generator. The ground-truth effect persists: every model scores 0.13–0.43 under PIX on v2, and the detector's PIX AUPRC (0.354) remains far below its BOX AUPRC. On all 3,066 v2 pairs the detector (seed 42) reaches 0.912 image AUROC, 0.915 on the primary set and 0.911 on pairs whose source document was seen in v1 training, so document overlap barely matters; the dataset authors report 0.599 for zero-shot TruFor and 0.585 for DocTamper, a reference rather than a like-for-like comparison, since our detector was trained on v1.

I. A patch classifier: the same shortcut in a third model family

The models above are trained to localize. To see whether the shortcut is specific to localization, we repeat the analysis with the image-level patch classifier of Sec. IV (Table VII). On the 400-image pilot subset (296 validation patches), 56 classical features (block DCT, LBP, GLCM, Laplacian residual, radial FFT) with an SVM and a random forest reach a patch AUROC of 0.646 and 0.706, a frozen ResNet-18 0.726, training from scratch 0.679 and full fine-tuning 0.899; the images are lossless PNG, so compression cues are absent, and only fine-tuning is carried forward. A pilot model trained on 400 images reaches a patch AUROC of 0.926 on the 1,431 validation patches; the unchanged recipe on the full training split reaches $0.983_{\pm 0.003}$ over three seeds. The decision rule matters as much as the data: with the uncalibrated rule "any patch ≥ 0.5" the full-data classifier flags 24 % of the authentic validation images, because the maximum over a median of 75 patches per image is biased upward; the thresholds selected on validation lie between 0.94 and 0.995 and bring the false-positive rate to 0.021 on validation and 0.022 on the 89 authentic test images (Wilson interval roughly 0.2–9 %), at a test AUROC of $0.978_{\pm 0.003}$. Averaging the top 2, 3 or 5 patches was not better than the maximum (validation AUROC 0.962, 0.958, 0.951 vs. 0.965). Under the same validation protocol the box U-Net reaches 0.921, 0.916 and 0.838 over its three seeds and the tiled detector 0.961.

TABLE VII

RESNET-18 PATCH CLASSIFIER. PATCH AUROC ON 1,431 VALIDATION PATCHES; IMAGE-LEVEL RESULTS WITH THE MAXIMUM PATCH SCORE AND THE VALIDATION-CALIBRATED THRESHOLD (TEST: 812 FORGED, 89 PAIRED AUTHENTIC). MEAN$\pm$SD OVER SEEDS 42–44 (SD OF TEST TPR AND FPR ≤ 0.017); THE PILOT HAS ONE SEED.

	Patch	Image AUROC		Test, calibrated	
Training data	AUROC	val	test	TPR	FPR
pilot, 400 images	0.926	0.893	0.919	0.805	0.079
full split	$0.983_{\pm 0.003}$	$0.965_{\pm 0.002}$	$0.978_{\pm 0.003}$	0.901	0.022
full, repaired	$0.982_{\pm 0.001}$	$0.948_{\pm 0.002}$	$0.958_{\pm 0.004}$	0.834	0.064

TABLE VIII

MARKER PROBES FOR THE PATCH CLASSIFIER AT ITS CALIBRATED THRESHOLD: DETECTION RATE ON THE 368 VALIDATION FORGERIES WITH RESIDUE BEFORE AND AFTER THE RESIDUE IS REPAIRED, FALSE-ALARM RATE ON THEIR AUTHENTIC COUNTERPARTS WITHOUT AND WITH THE RESIDUE PASTED IN, AND TEST DETECTION RATE WITH AND WITHOUT RESIDUE. THREE-SEED MEANS (SD ≤ 0.047); PILOT: ONE SEED.

	forged TPR		authentic FPR		test TPR	
Training data	raw	repaired	raw	+marker	residue	none
pilot, 400 images	0.910	0.207	0.073	0.848	0.906	0.465
full split	0.995	0.471	0.016	0.955	0.985	0.618
full, repaired	0.880	0.899	0.049	0.458	0.863	0.735

The headline numbers are, however, largely a marker score (Table VIII). When only the residue pixels are restored, the full-data classifier misses half of the forgeries it had found (detection rate 0.995 to 0.471), and pasting only the residue onto authentic images makes it flag 96 % of them. On the test set it detects 98.5 % of the 627 forgeries with residue but 61.8 % of the 185 without, and 62.4 % of the 85 Ideogram edits. Training on marker-repaired patches removes most of the dependence (0.880 raw vs. 0.899 repaired) and gives the best detection of residue-free and Ideogram forgeries (0.735

and 0.737), at the price of a lower overall AUROC (0.958) and a higher false-alarm rate (0.064). The repaired model still flags 46 % of the authentic images that carry a pasted residue; it never saw the marker during training, and the pasted pixels are themselves a manipulation. The shortcut found for the segmenters and, more weakly, for the detector thus reappears in a third model family and with a different output type. These classifiers were not evaluated on v2, under degradations or with a held-out generator.

J. Attribution evidence

Input interventions show what a model reacts to; attributions show where its decision comes from. We compute SHAP values [63] with the model-agnostic partition explainer (200 evaluations, window-score target) for the three standard U-Nets and the pilot classifier on eight inputs (three true positives, three false negatives, two true negatives). Additivity holds exactly in every run. Deleting regions in the order of their attribution lowers the score faster than a random order [64] (area over the perturbation curve 0.445 vs. 0.191 for the box U-Net, 0.321 vs. 0.101 for PIX, 0.349 vs. 0.167 for GLYPH, 0.623 vs. 0.225 for the classifier), so the maps point at regions the models use. The positive attribution mass on residue pixels is 4.3 (box U-Net), 8.6 (PIX), 7.5 (GLYPH) and 15.1 (classifier) times their share of the image area, and 4.1–6.2 times on a ring around the box edge. With eight inputs and a coarse partition this is qualitative support for Sec. V-C, not an independent measurement. An interactive demonstrator that runs the models, probes and explanations on dataset samples accompanies the code.

VI. Discussion

What the box label rewards. Three independent lines of evidence point the same way. The same predictions lose 0.55 AUPRC when scored against the pixels that actually changed (Sec. V-A); the box model is indifferent to blurring the box interior but not its rim; and a box detector trained on the same labels reaches 0.969 test AUPRC under the box metric, more than any segmenter, while scoring 0.368 under the pixel-change truth. A benchmark that scores localization against the edited field's box therefore measures *field detection*. That is a useful capability for triage, and box detection delivers it with the highest box scores, but it should not be reported as pixel-level tamper localization.

The marker is a shortcut, not the whole story. Pasting only the residual green line onto an authentic image is enough for the standard box model to find the field, and its test AP drops from 0.97 on images that carry the residue to 0.69 on images that do not. Yet box models retrained on marker-repaired images (three seeds) keep 0.912 validation AUPRC (0.923 before) and score similarly on residue-free Gemini images (0.762 vs. 0.722, overlapping CIs). The residue is used when present and replaced by other cues when removed, which means the headline AIForge-Doc numbers overstate the difficulty-adjusted performance on marker-bearing images, but do not consist of the marker alone. Since the residue is present in 86 % of Gemini and 2 % of Ideogram forgeries, held-out-generator results on v1 confound generator and marker; repairing the marker is a prerequisite for any generator-transfer claim on this dataset. The dependence is not specific to localizers: an image-level patch classifier loses half of its detections when the residue is repaired, and attributions of all four examined models concentrate on the residue pixels (Secs. V-I and V-J). The marker-free v2 benchmark confirms this from outside: models trained with the marker lose most on exactly the specifications that carried it, and the repaired box model loses nothing on the others (Sec. V-H).

Recommendations for document-forgery benchmarks. (i) Release the authentic source with every training forgery, as AIForge-Doc does, and report a pixel-change metric next to the box metric. (ii) Remove generation-pipeline annotations (markers, crop borders) from the released images and audit residue with a simple colour test. (iii) Report box-level AP separately from pixel metrics, with a box detector as a baseline. (iv) Report leakage-free test subsets and group-aware confidence intervals.

Limitations. The pixel-change ground truth is derived automatically (ε=8 with morphology). A vision-language check rated it OK in 69 of 96 audited pairs, but a stricter human audit of the same stratified 100 pairs by the author rated it partial in 64 and wrong in none: in all 64 the auditor's note describes over-coverage, the mask also marking re-rendered background or unchanged content of the field (WildReceipt 39/39 partial, XFUND 0/11), and the two audits agree poorly (Cohen's κ=0.07, 95 % CI $[-0.08, 0.21]$). The same audit found the official box correct in 99 of 100 pairs. PIX therefore over-covers rather than misses the edit; the human audit has a single annotator, the author, so no inter-annotator agreement is available. The second architecture, the marker-repaired pixel and glyph models, the joint glyph model, the 20-epoch runs and the YOLO ablations use one seed; the main U-Net grid, the marker-repaired box model, the joint pixel model, the held-out-generator U-Nets, the main YOLO model and the patch classifier use three. Ideogram contributes 126 held-out images. TruFor could not be fine-tuned because its training code is not released, so learned in-domain models are compared with zero-shot references only. Image-level detection is measured on validation pairs because the test authentic set is not paired. The patch classifier is the exception: its threshold is calibrated on validation pairs and tested against the 89 paired authentic test images, so its test false-positive rate has a wide interval; its thresholds lie close to saturation (0.94–0.995) and vary with the seed; it was not evaluated on v2, under degradations or with a held-out generator. The attribution analysis covers eight inputs at a low evaluation budget. The external test on AIForge-Doc v2 uses models trained on v1 only; the marker-repaired pixel and glyph models there have one seed, and v2 changes both the generator and the paste-back procedure, which our same-specification comparison separates only through the v1 residue split.

VII. Conclusion

On AI-inpainted financial documents, the official box masks train and reward detectors of the edited field, not localizers of the changed pixels: changing only the training mask reverses the model ranking, the effect replicates across decoders, and a tiled box detector is the best model under the box metric and transfers best across generators, yet localizes changed pixels no better than the box-trained U-Net. A residual edit marker in most Gemini forgeries is exploited when present, by the segmenters, more weakly by the detector and most strongly by an image-level patch classifier; retraining without it preserves most of the box score but removes the marker-specific advantage. Pixel-change supervision gives the most faithful masks and the fewest false alarms, but it does not close the generator gap (joint supervision was not tested on the held-out generator). On the marker-free AIForge-Doc v2, the segmenters trained on marker-bearing v1 images lose 43–64 % of their box AUPRC, the marker-repaired box model (24 %) and the tiled detector (8 %) far less. We recommend reporting pixel-change and box metrics side by side, auditing generation residue, and treating held-out-generator results on marker-bearing data with caution.

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